Introduction to Credit Derivatives

Seminar Module

conducted by

ACF Consultants Ltd





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Topics Covered

- Buying and selling credit risk
- Credit derivatives as unbundling tools
- Credit derivatives as a means to transfer credit risk
- The credit derivatives market
- Types of credit derivative

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Key Point Summary

- Credit derivatives allow the transfer of credit risk from a Protection Buyer to a Protection Seller
- In exchange for taking on the credit risk, the protection seller receives a fee payment
- Credit derivatives allow for the unbundling and pricing of credit risk
- Credit derivatives can be used for a wide set of credit-related applications, not just for protecting against default
- Reference entities cover a wide spectrum, but trading volume is dominated by trades on sovereigns and financial entities
- The most important credit derivatives are single-name credit default swaps (CDS) and credit indices

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Buying and Selling Credit Risk

Credit Derivative transactions generally involve two counterparties:

- One party is the **Protection Buyer** (Credit Risk Seller)
- The other party is the **Protection Seller** (Credit Risk Buyer)

The Protection Buyer transfers **credit risk** to the protection seller.

One way to do this is through a Credit Default Swap…



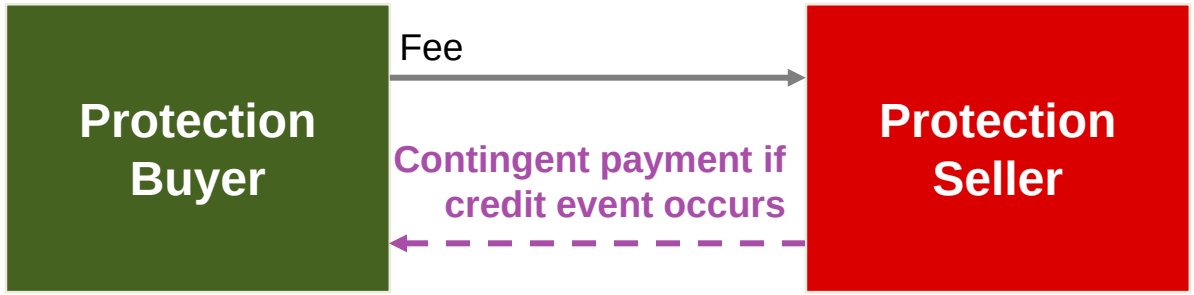
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Credit Default Swaps

A Credit Default swap involves two counterparties:

- The Protection Buyer:
  - Pays a fixed annual fee
  - Receives a contingent payment in the event of a credit event on the reference entity
- The Protection Seller:
  - Receives the fixed annual fee
  - Makes a contingent payment in the event of a credit event on the reference entity

- Credit Default Swaps only pay out if a recognised credit event occurs...
- ... but the value of an existing contract will change as market conditions change

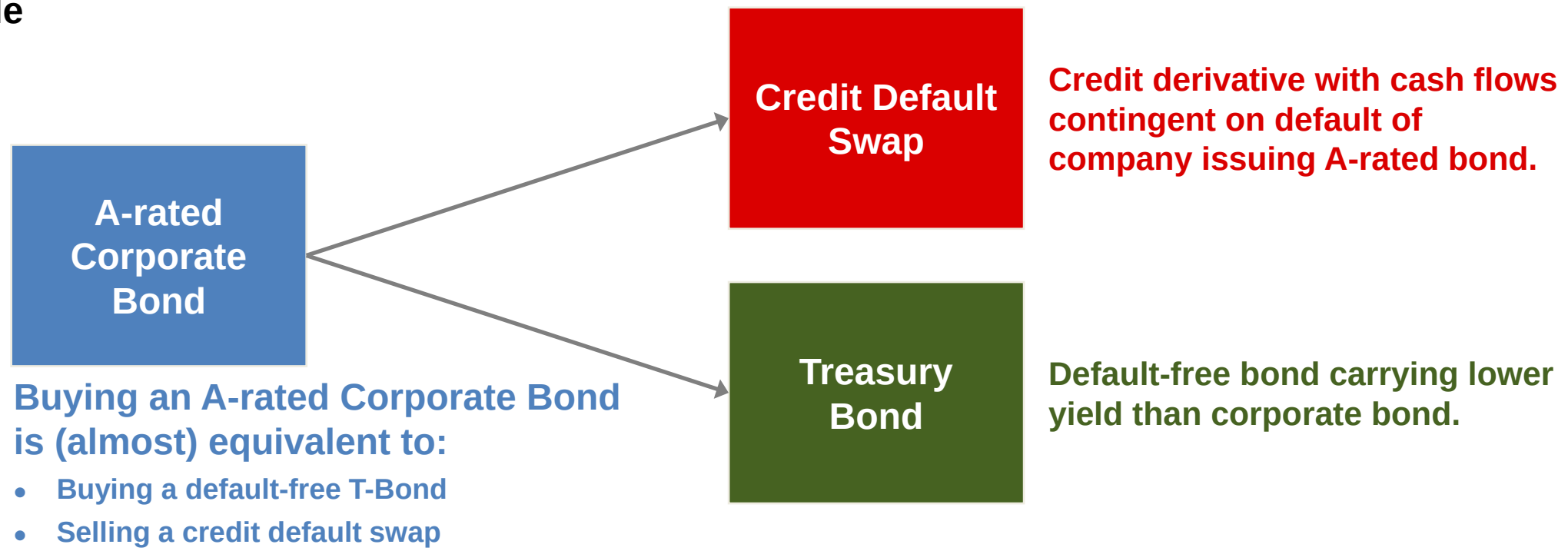


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Unbundling Credit Risk

- Credit Derivatives:
  - Unbundle credit risk from other financial instruments
  - Allow users to trade and manage credit risk separately from other risks and cash flows

Example

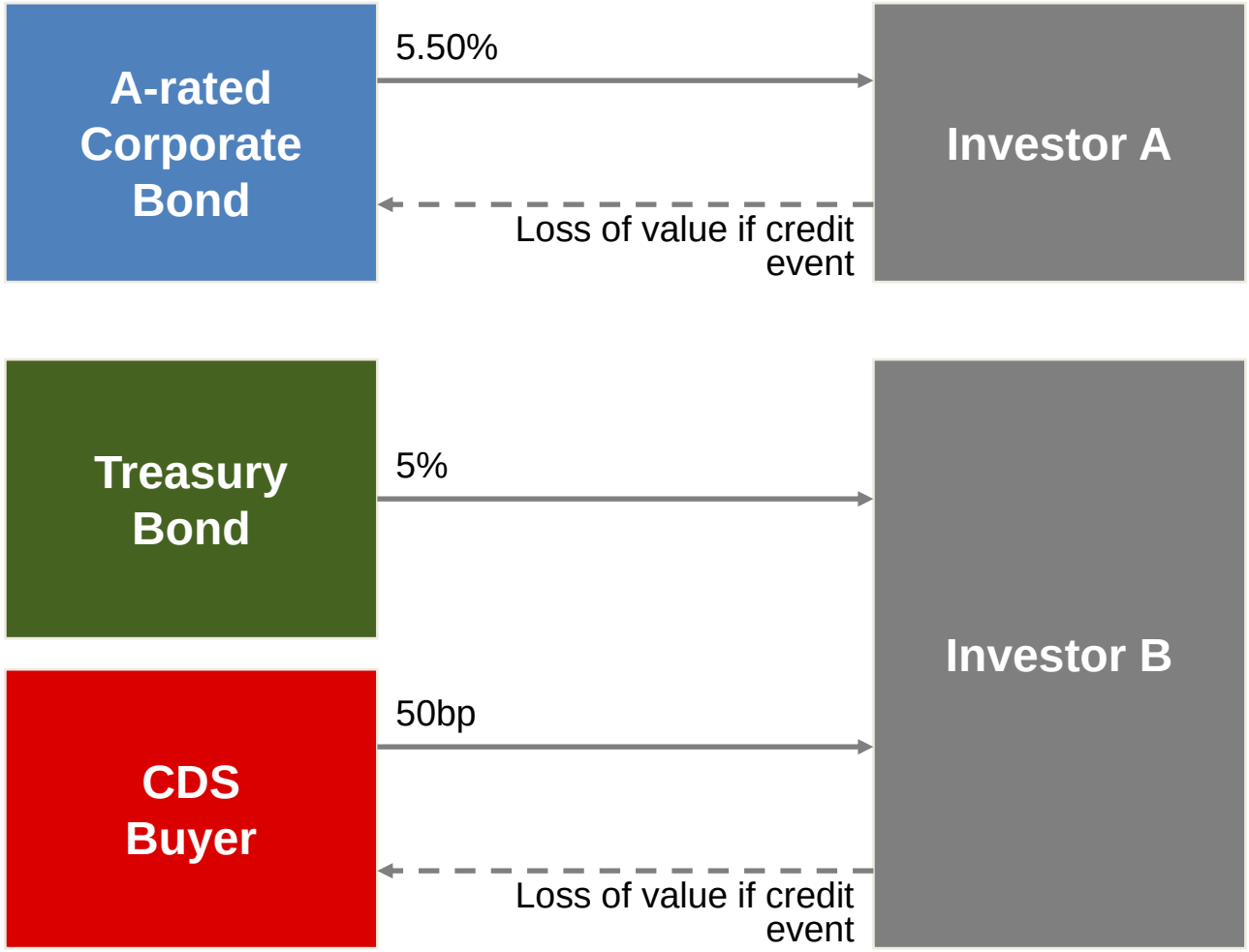


Buying an A-rated Corporate Bond is (almost) equivalent to:

- **Buying a default-free T-Bond**
- **Selling a credit default swap**

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Unbundling Credit Risk



Investor A owns an A-rated corporate bond.

Investor B owns a T-Bond and sells a credit default swap on the A-rated bond.

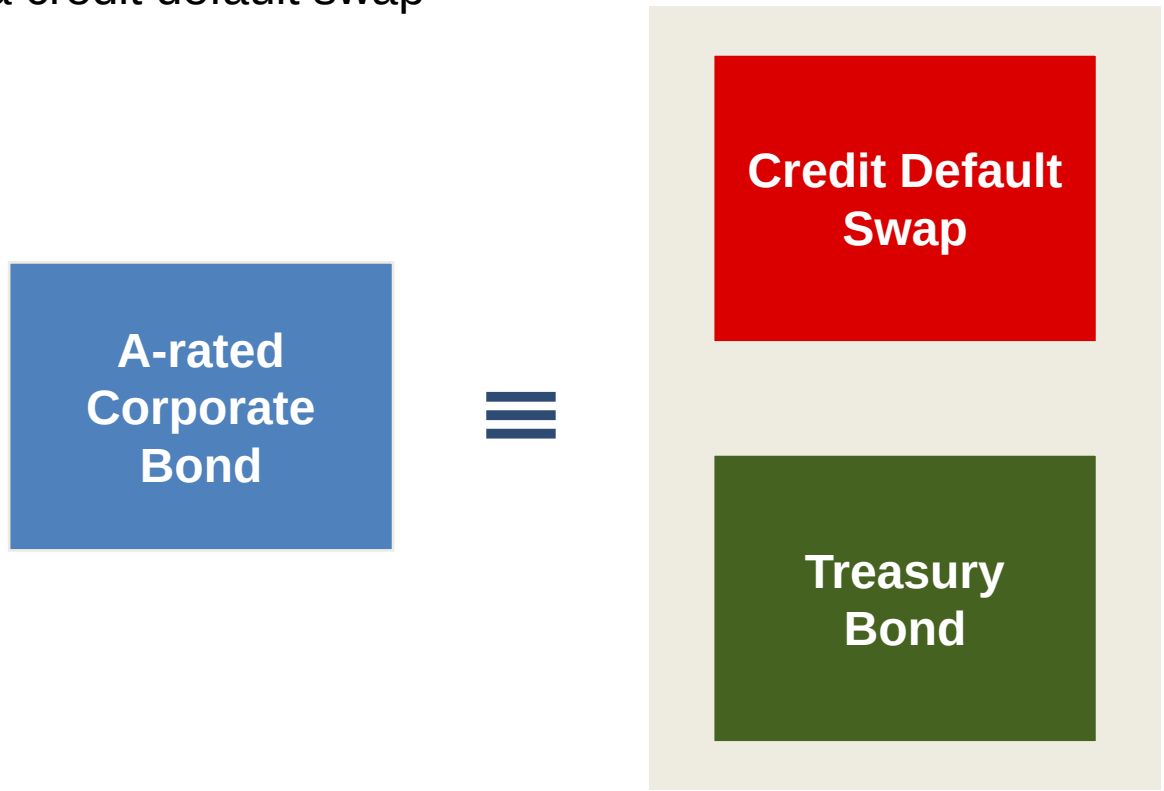
Investor receives 50bp p.a. from the CDS, but must compensate the holder of the CDS in the event of credit losses on the A-rated bond.

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Unbundling Credit Risk

An A-rated bond can therefore be considered as:

- Long a default-free bond
- Short a credit default swap



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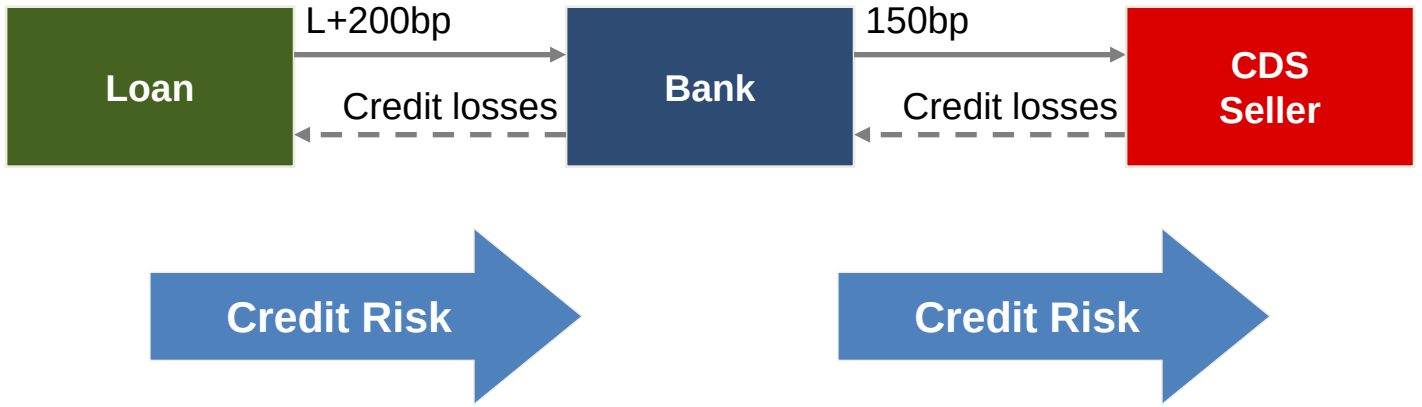
Motivations for Using Credit Derivatives

- To **manage credit risk**
  - Credit risk may now more easily be transferred from one party to the next
- To enable users to **diversify credit risk**
  - Parties can selectively increase or decrease exposure to:
    - Countries
    - Market sectors
    - Types of financial instruments
- To **earn income**
  - Low-cost borrowers with large balance sheets can earn income from parties who want credit exposure without owning assets
- To provide **access to exposures** that would not otherwise be available
  - e.g. Investors can gain access to syndicated loans
- To **trade a view** on credit
  - The credit risk of an entity (or the market) will improve
  - The credit risk of an entity (or the market) will worsen
  - A credit event (like a default or a bankruptcy) may happen

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Simple Example – Bank Sells Credit Risk

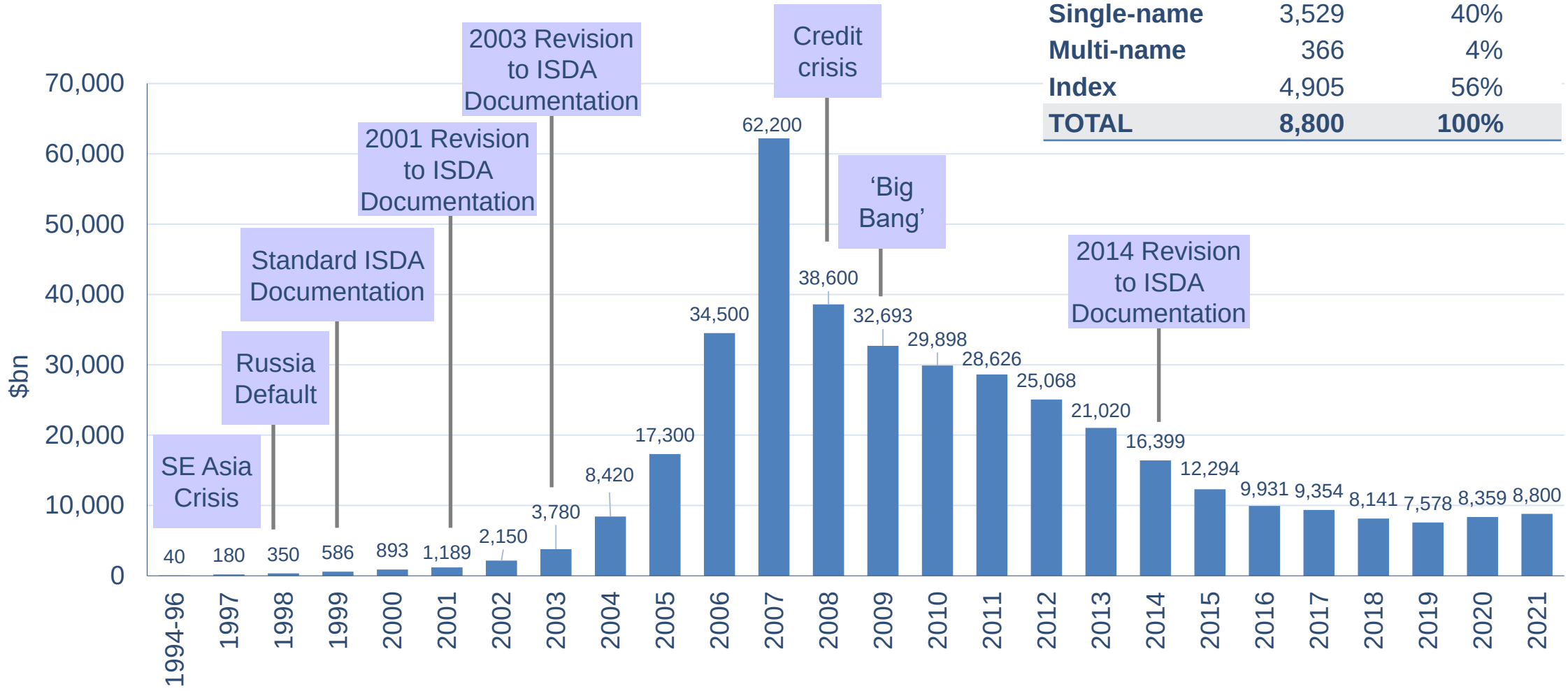
- Bank loans are exposed to credit risk
- Part of the interest charged is intended to cover the cost of bad debts
- A bank could buy a credit default swap:
  - Bank pays a regular payment to the credit swap counterparty
  - Credit swap counterparty makes good any credit losses suffered by bank on protected loan
- Bank **buys** credit protection and **sells** credit risk
- Net return for bank is now guaranteed, irrespective of bad debt rates



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Credit Derivatives Market Size

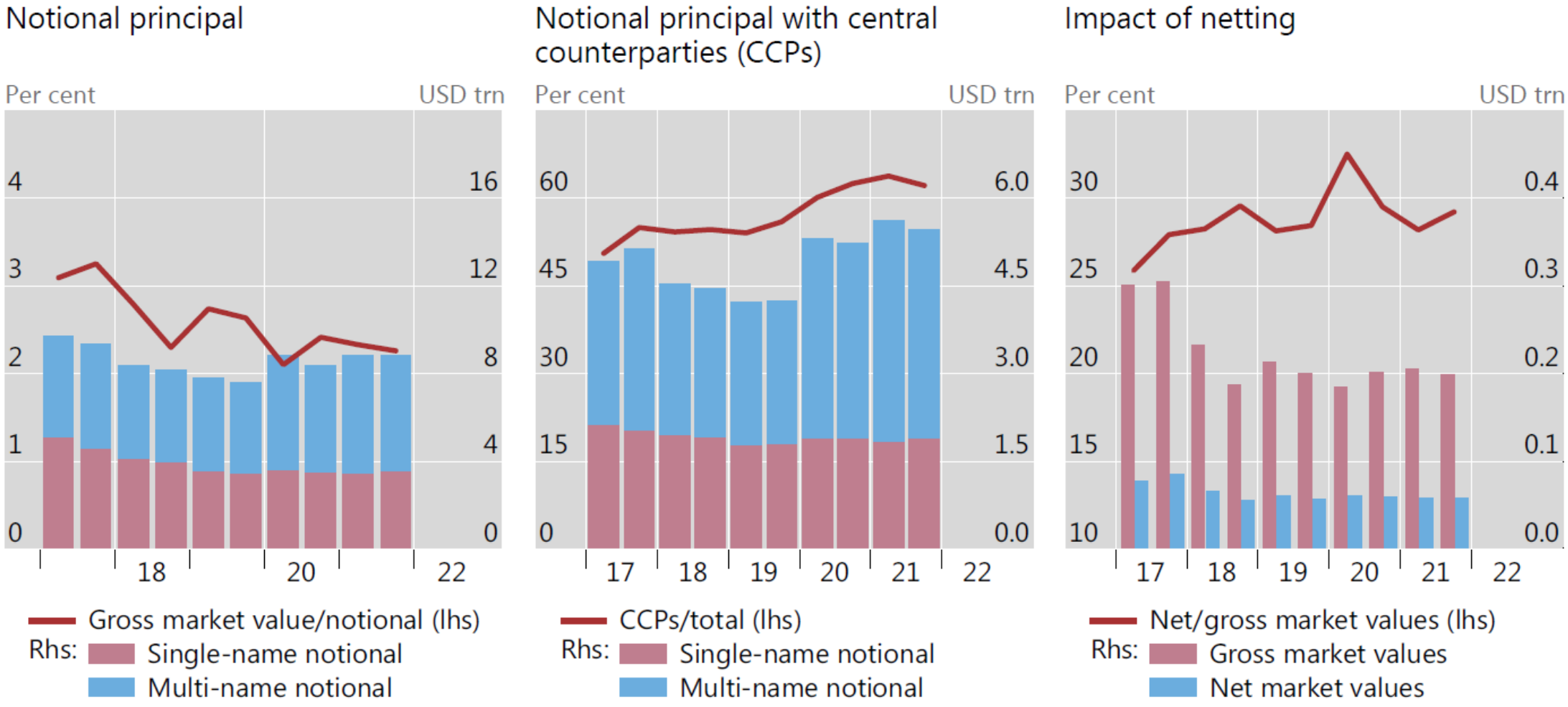
	$ trn	% age
Single-name	3,529	40%
Multi-name	366	4%
Index	4,905	56%
TOTAL	**8,800**	**100%**



Credit derivatives in this chart comprise credit default swaps referencing single names, indexes, baskets, and portfolios.
Source: British Bankers Association Credit Derivatives Survey 2006; ISDA Market Surveys 2001 to 2008, BIS 2009 onwards

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Credit Derivatives Market Size – Gross vs. Net

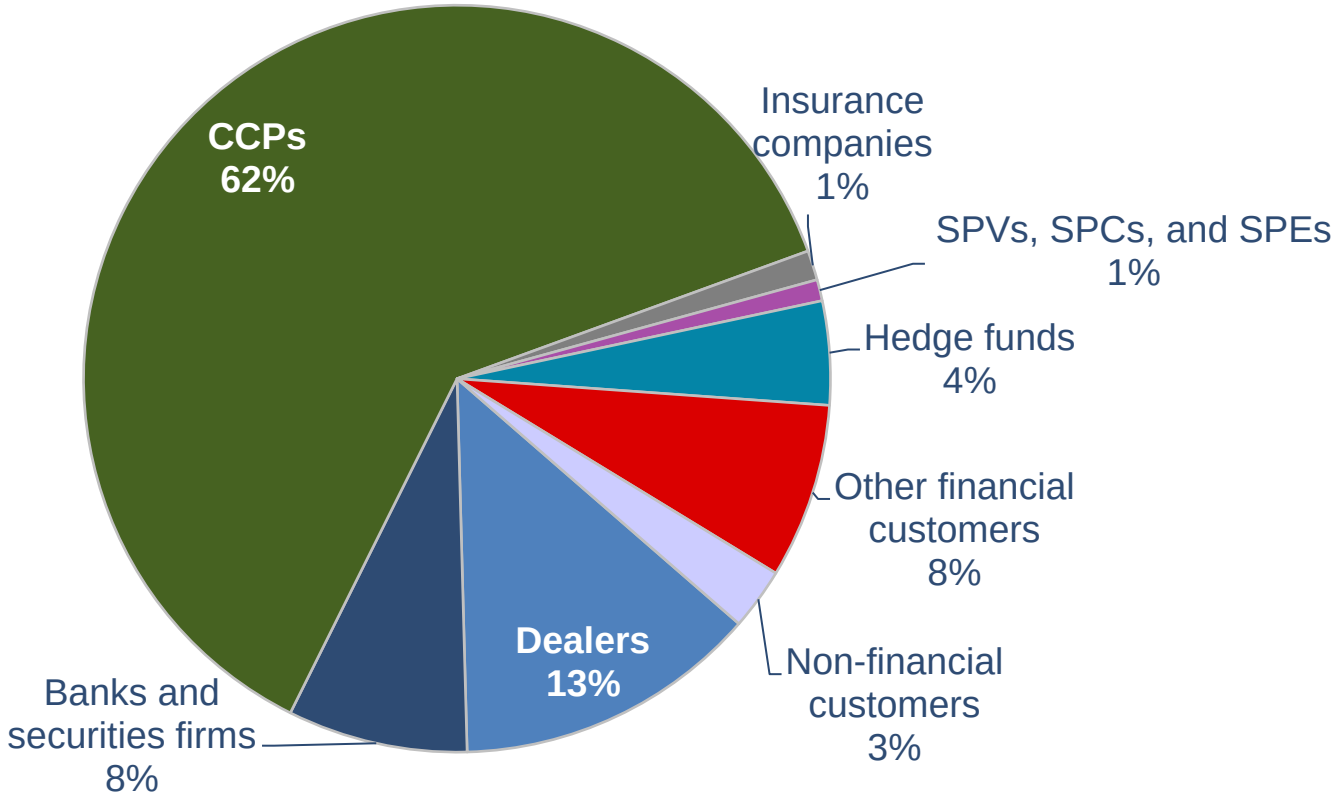


Source: OTC Derivatives Statistics at end-December 2021

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Analysis of Credit Derivatives by Counterparty

(Based on Notional Amounts Outstanding)

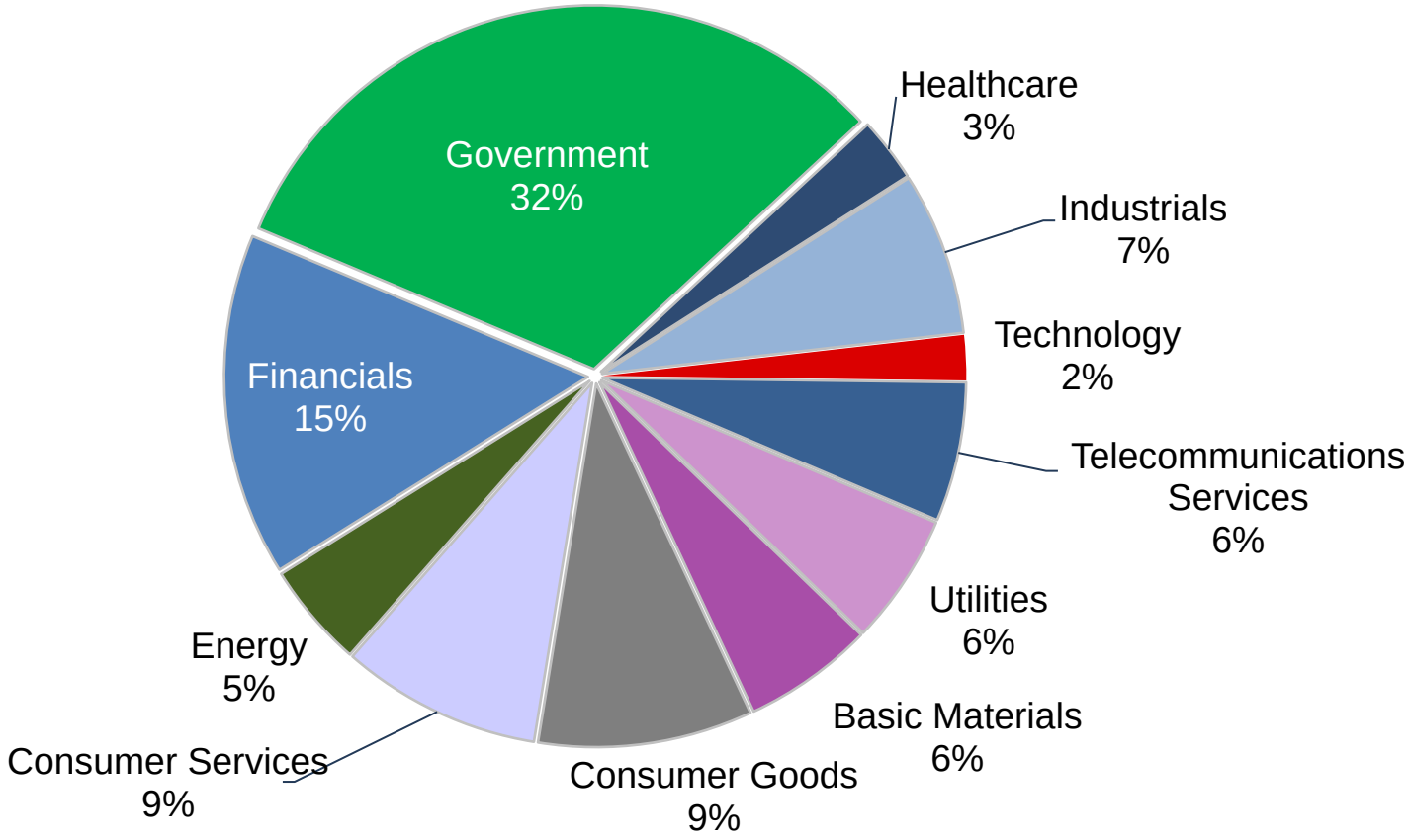


Source: OTC Derivatives Statistics at end-December 2021

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Analysis of Credit Derivatives Reference Entities

By Average Notional Traded, 27 Aug 2022 to 24 Feb 2023



Source: DTCC Average Weekly Data by Reference Entity

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Uses of Credit Derivatives

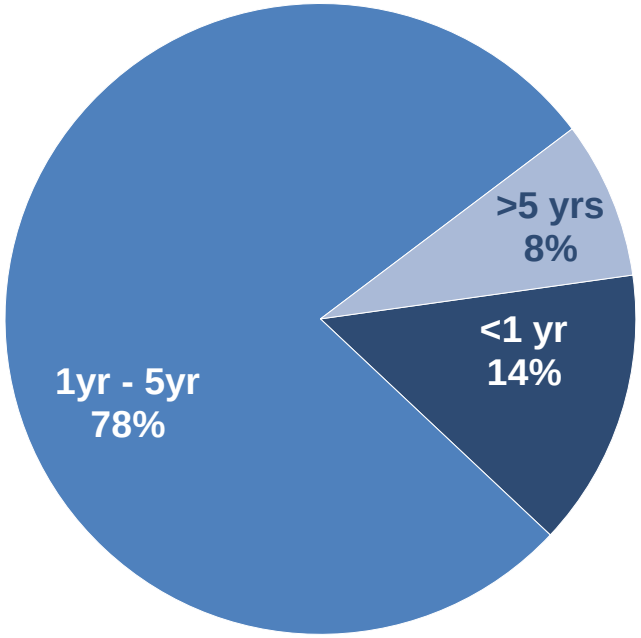
	Reasons They Buy Protection	Reasons They Sell Protection
Banks	• Credit risk management • Regulatory capital relief	• Geographical/industry diversification of loan portfolio • Offset costs of hedging other credits • Enhance yield versus lending
Hedge Funds	• Express negative views on a credit • Package with bonds for basis/convexity trades • Package with converts to isolate equity optionality • Put on forward long or short positions	• Express positive views on a credit ▪ Can be cheaper than bonds ▪ Inherent leverage • Put on forward long or short positions
Insurance Companies	• Reduce or diversify liability concentrations in the insurance portfolio • Create forward long positions that allow them to hedge near-term default risk	• Increase diversification • Enhance yield (cheaper alternative to bonds)
Asset Managers	• Strategic trade construction • Put on forward trades • Express negative views on a credit (only certain funds)	• Express positive views on a credit • Yield enhancement (can be cheaper than bonds) • Increase diversification
Securities Firms	• Market intermediary • Credit risk management • Regulatory capital relief	• Market intermediary • Geographical/industry diversification of loan portfolio • Offset costs of hedging other credits

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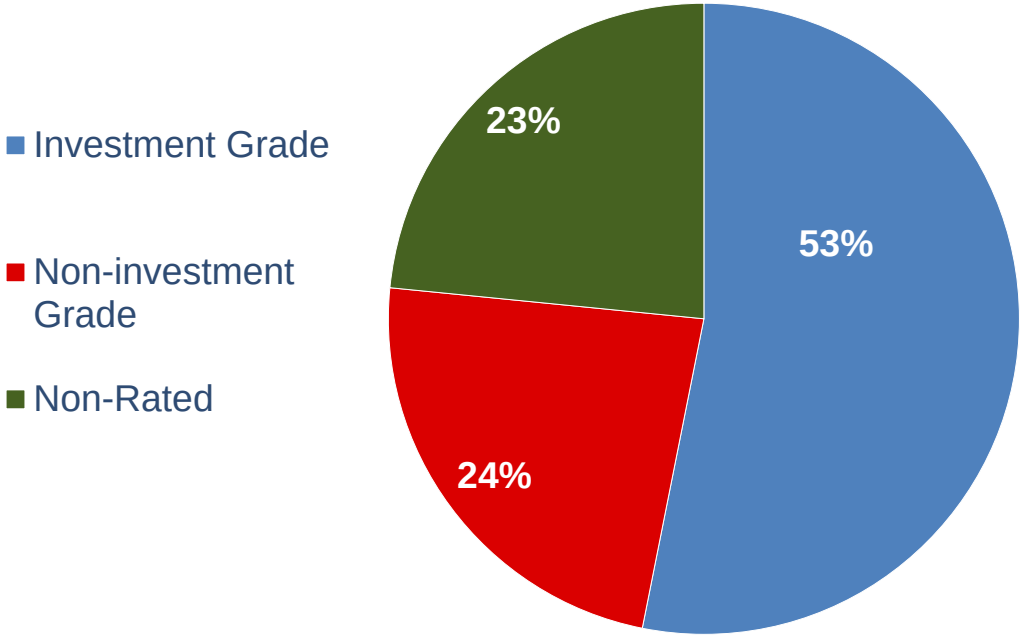
Maturity and Credit Analysis

By Notional Outstanding

Maturity



Rating category



Source: BIS Statistics Explorer at end-December 2021

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Types of Credit Derivative

- Credit Default Swaps (CDS)
  - Single-name
  - Basket
- Index Products
- Collateralised products:
  - CDOs – Collateralised Debt Obligations
  - CBOs – Collateralised Bond Obligations
  - CLOs – Collateralised Loan Obligations
- Credit Default Swaptions
- Credit-Linked Notes (CLN)
- Credit Spread Options (CSO)
- Total Return Swaps (TRS)

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